

A TWO-GRID FINITE-ELEMENT METHOD FOR THE NONLINEAR SCHRÖDINGER EQUATION*

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Abstract

In this paper, some two-grid finite element schemes are constructed for solving the nonlinear Schrödinger equation. With these schemes, the solution of the original problem is reduced to the solution of the same problem on a much coarser grid together with the solutions of two linear problems on a fine grid. We have shown, both theoretically and numerically, that our schemes are efficient and achieve asymptotically optimal accuracy.

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1. Introduction

Nonlinear Schrödinger equations arise from mathematical modelling of problems in various areas such as fluid dynamics, nonlinear optics, plasma physics, protein chemistry, etc., see, e.g., [2, 12, 13]. In this paper, we will study two-grid finite element discretization schemes for the boundary value problem of the nonlinear Schrödinger equation:

$$-\Delta\psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x}) + |\psi(\mathbf{x})|^2\psi(\mathbf{x}) = f(\mathbf{x}), \quad \forall \mathbf{x} \in \Omega, \quad (1.1)$$

$$\psi(\mathbf{x}) = 0, \quad \forall \mathbf{x} \in \partial\Omega, \quad (1.2)$$

where $\Omega \subset R^2$ is a convex polygonal domain, $f(\mathbf{x}), V(\mathbf{x})$ and unknown function $\psi(\mathbf{x})$ are complex-valued. For any complex-valued function w , we denote its real part by w_1 , the imaginary part by w_2 . Then problem (1.1)-(1.2) is equivalent to the following coupled nonlinear equations:

$$-\Delta\psi_1(\mathbf{x}) + V_1(\mathbf{x})\psi_1(\mathbf{x}) + \psi_1^3(\mathbf{x}) + \psi_2^2(\mathbf{x})\psi_1(\mathbf{x}) - V_2(\mathbf{x})\psi_2(\mathbf{x}) = f_1(\mathbf{x}), \quad \forall \mathbf{x} \in \Omega, \quad (1.3)$$

$$-\Delta\psi_2(\mathbf{x}) + V_1(\mathbf{x})\psi_2(\mathbf{x}) + \psi_2^3(\mathbf{x}) + \psi_1^2(\mathbf{x})\psi_2(\mathbf{x}) + V_2(\mathbf{x})\psi_1(\mathbf{x}) = f_2(\mathbf{x}), \quad \forall \mathbf{x} \in \Omega, \quad (1.4)$$

$$\psi_j(\mathbf{x}) = 0, j = 1, 2, \quad \forall \mathbf{x} \in \partial\Omega. \quad (1.5)$$

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The two-grid discretization method, proposed originally by Xu [17] in 1992, is an efficient numerical method. Later, Xu [18, 19] introduced the two-grid finite element approach to solve nonlinear elliptic equations efficiently, where the basic idea is to use a coarse space to produce a rough approximation of the solution, and then use it as the initial guess for one Newton iteration on the fine grid. This procedure involves a nonlinear solver on coarse space and a linear solver on fine space. Now the idea of the two-grid discretization method has already been applied to solving many problems, such as nonlinear parabolic equations [5], nonlinear elasticity problems [1], two-phase mixed-domain PEMFC model [9], and the Schrödinger equations [3, 6–8, 10, 11, 15, 16, 20].

In the literature, there have been only a few papers on two-grid discretization methods for Schrödinger equation. Ignat et al. [10] constructed two-grid finite difference scheme for nonlinear Schrödinger equations, where the equations on the fine grid are linearized, but not decoupled. Jin et al. [11] successfully extended the two-grid finite element method to solve coupled partial differential equations such as linear Schrödinger equation, where the equations on fine grid are decoupled, so that the computational complexity of solving Schrödinger equation is comparable to solving two decoupled Poisson equations on the same fine grid. Zhou et al. [7, 8] proposed a two-scale finite element discretization scheme for eigenvalue problem of Schrödinger equation. The approach is a powerful technique in obtaining accurate and efficient approximations for large scale quantum eigenvalue problems. Chang et al. [3] combined the two-grid discretization together with the predictor-corrector method and developed an algorithm for computing the extremum eigenpairs of the discrete Schrödinger eigenvalue problem. Chien et al. [6] proposed two-grid discretization schemes with two-loop continuation algorithms for nonlinear Schrödinger equations, where the centered difference approximations, the six-node triangular elements and the Adini elements are used to discretize the PDEs. Numerical experiments have shown that these schemes were efficient, but no rigorous error analysis were given. Wu [15, 16] constructed two-grid mixed finite element schemes for nonlinear Schrödinger equations, where a linear and indefinite (the typical nature of mixed finite element) discretization systems are solved on the fine grid. Numerical experiments using these schemes have been shown to be efficient, however, no rigorous error analysis has been conducted. Recently, Zhang et al. [20] extended the approach given in [11] to time-dependent linear Schrödinger equation.

In this paper, we follow the idea of [11, 20] to apply two-grid finite element method to solve the nonlinear problem (1.3)-(1.5). Specifically, we use Newton iteration method to solve the original problem directly on coarse grid, and algebraic multigrid method to solve the discrete systems of the linearized and decoupled equations on fine grid. The resulting solution, verified by theoretical analysis and numerical experiments, achieves optimal accuracy in H^1 -norm.

The rest of the paper is organized as follows: Section 2 is a description and analysis of the finite element method for Schrödinger equation. In Section 3, we construct the two-grid finite element schemes and derive the error estimates. In Section 4, we demonstrate numerical examples to verify the efficiency and effectiveness of the schemes.

2. The Finite Element Approximation

For any real-valued and Lebesgue square integrable functions $\varphi_1(\mathbf{x})$ and $\varphi_2(\mathbf{x})$, let (φ_1, φ_2) denote the inner product

$$(\varphi_1, \varphi_2) = \int_{\Omega} \varphi_1(\mathbf{x}) \varphi_2(\mathbf{x}) d\mathbf{x}.$$

Also, for any real-valued function $\varphi(\mathbf{x})$ and integer $m \geq 0$, let $W_p^m(\Omega)$ be the standard Sobolev space with norms given by

$$\|\varphi\|_{m,p} = \left(\sum_{|\alpha| \leq m} \|D^\alpha \varphi\|_{L^p(\Omega)}^p \right)^{\frac{1}{p}}, \quad 1 \leq p < \infty,$$

and $\|\varphi\|_{m,\infty} = \max_{|\alpha| \leq m} \|D^\alpha \varphi\|_{L^\infty(\Omega)}$. For simplification, we denote W_2^m by H^m , $\|\cdot\|_{m,2}$ by $\|\cdot\|_m$, and the subspace of $H^1(\Omega)$ consisting of functions with vanishing trace on $\partial\Omega$ by $H_0^1(\Omega)$.

For any complex-valued function w , let

$$\|w\|_{m,p} = (\|w_1\|_{m,p}^p + \|w_2\|_{m,p}^p)^{\frac{1}{p}}, \quad \|w\|_{m,\infty} = \|w_1\|_{m,\infty} + \|w_2\|_{m,\infty}.$$

Then $\psi(\mathbf{x})$, the weak solution of problem (1.1)-(1.2), is defined as follows: Find $\psi(\mathbf{x}) \in H_0^1 \times H_0^1$ such that

$$A(\psi, \chi) = (f, \chi), \quad \forall \chi \in H_0^1 \times H_0^1,$$

where

$$\begin{aligned} A(\psi, \chi) &= (\nabla \psi_1, \nabla \chi_1) + (V_1 \psi_1 + \psi_1^3 + \psi_2^2 \psi_1 - V_2 \psi_2, \chi_1) \\ &\quad + (\nabla \psi_2, \nabla \chi_2) + (V_1 \psi_2 + \psi_2^3 + \psi_1^2 \psi_2 + V_2 \psi_1, \chi_2), \\ (f, \chi) &= (f_1, \chi_1) + (f_2, \chi_2). \end{aligned}$$

Let T_h be a quasi-uniform triangulation of Ω with mesh size $h > 0$, $S_0^h \subset H_0^1$ be the corresponding piecewise linear finite element space, and throughout this paper C denotes a positive constant independent of h , but may have different value at different place. We also assume that h is small enough such that for the constant C given in (2.31), the following inequality holds:

$$Ch < \alpha < 1, \quad (2.1)$$

where α is a constant.

Then $\psi_h(\mathbf{x})$, the finite element approximation of problem (1.1)-(1.2), is defined as the following: Find $\psi_h(\mathbf{x}) \in S_0^h \times S_0^h$ such that

$$A(\psi_h, \chi_h) = (f, \chi_h), \quad \forall \chi_h \in S_0^h \times S_0^h. \quad (2.2)$$

Let $u_I \in S_0^h \times S_0^h$ be the linear interpolation of u , then u_I satisfies the following error estimation:

$$|u - u_I|_{l,q,\Omega} \leq Ch^{2-l+\frac{2}{q}-\frac{2}{p}} \|u\|_{2,p,\Omega}, \quad p \leq q, l = 0, 1. \quad (2.3)$$

Our basic assumption is that $\psi(\mathbf{x})$, the unique solution of problem (1.1)-(1.2), satisfies $\psi(\mathbf{x}) \in (H^2 \cap H_0^1) \times (H^2 \cap H_0^1)$. Introducing the bilinear form

$$\begin{aligned} A'(\phi; u, \chi) &= (\nabla u_1, \nabla \chi_1) + (V_1 u_1 + 3\phi_1^2 u_1 + \phi_2^2 u_1 - V_2 u_2 + 2\phi_1 \phi_2 u_2, \chi_1) \\ &\quad + (\nabla u_2, \nabla \chi_2) + (V_1 u_2 + \phi_1^2 u_2 + 3\phi_2^2 u_2 + V_2 u_1 + 2\phi_1 \phi_2 u_1, \chi_2) \end{aligned}$$

and the nonlinear form

$$\begin{aligned} R(\phi, u, \chi) &= (3\phi_1(\phi_1 - u_1)^2 + 2\phi_2(\phi_1 - u_1)(\phi_2 - u_2) + \phi_1(\phi_2 - u_2)^2 - (\phi_1 - u_1)^3 \\ &\quad - (\phi_1 - u_1)(\phi_2 - u_2)^2, \chi_1) + (3\phi_2(\phi_2 - u_2)^2 + 2\phi_1(\phi_2 - u_2)(\phi_1 - u_1) \\ &\quad + \phi_2(\phi_1 - u_1)^2 - (\phi_2 - u_2)^3 - (\phi_2 - u_2)(\phi_1 - u_1)^2, \chi_2), \end{aligned} \quad (2.4)$$

we have the lemmas as follows.

Lemma 2.1. For any $v, \omega, \chi \in H_0^1 \times H_0^1$,

$$A(\omega, \chi) = A(v, \chi) + A'(v; \omega - v, \chi) + R(v, \omega, \chi). \quad (2.5)$$

Thus $\psi_h \in S_0^h \times S_0^h$ solves (2.2) if and only if

$$A'(\psi; \psi - \psi_h, \chi_h) = R(\psi, \psi_h, \chi_h), \quad \forall \chi_h \in S_0^h \times S_0^h. \quad (2.6)$$

Proof. Set $\eta(t) = A(v + t(\omega - v), \chi)$, then (2.5) follows from the elementary identity

$$\eta(1) = \eta(0) + \eta'(0) + \int_0^1 \eta''(t)(1-t)dt.$$

A straightforward calculation shows that

$$\eta'(0) = A'(v; \omega - v, \chi), \quad R(v, \omega, \chi) = \int_0^1 \eta''(t)(1-t)dt.$$

Therefore, (2.6) follows from (2.2) and (2.5). $\square$

Next, we consider the problem: for given ϕ and g , find $u \in H_0^1 \times H_0^1$ such that

$$A'(\phi; u, \chi) = (g, \chi), \quad \forall \chi \in H_0^1 \times H_0^1. \quad (2.7)$$

We have the following regularity result:

Lemma 2.2. Assume that $g(x) \in L^2(\Omega) \times L^2(\Omega)$, $V(x), \phi(x) \in L^\infty(\Omega) \times L^\infty(\Omega)$ and $V_1(x) \geq 0$, then the problem (2.7) has a unique solution $u \in H^2 \times H^2$ satisfying

$$\|u\|_2 \leq C\|g\|_0. \quad (2.8)$$

Proof. For any $\omega, \chi \in H_0^1 \times H_0^1$, we can easily check that

$$|A'(\phi; \omega, \chi)| \leq C\|\omega\|_1\|\chi\|_1, \quad A'(\phi; \omega, \omega) \geq C\|\omega\|_1^2. \quad (2.9)$$

Therefore, by the Lax-Milgram theorem, the variation problem (2.7) has a unique solution $u \in H_0^1 \times H_0^1$. Moreover, by the regularity theory for elliptic boundary value problems, we have

$$\begin{aligned} \|u_1\|_2 &\leq C\|g_1 + V_2 u_2 - 2\phi_1 \phi_2 u_2\|_0 \leq C(\|u_2\|_0 + \|g_1\|_0), \\ \|u_2\|_2 &\leq C\|g_2 - V_2 u_1 - 2\phi_1 \phi_2 u_1\|_0 \leq C(\|u_1\|_0 + \|g_2\|_0), \end{aligned}$$

the above two inequalities imply that

$$\|u\|_2 \leq C(\|u\|_0 + \|g\|_0). \quad (2.10)$$

From (2.7) and (2.9), we have

$$C\|u\|_1^2 \leq A'(\phi; u, u) = (g, u) \leq \|g\|_0\|u\|_0,$$

which implies that

$$\|u\|_1 \leq C\|g\|_0. \quad (2.11)$$

Therefore, (2.8) follows from (2.10) and (2.11). $\square$

Now, we define the elliptic projection operator $P_h : H_0^1 \times H_0^1 \longrightarrow S_0^h \times S_0^h$ by

$$A'(\phi; P_h u, \chi_h) = A'(\phi; u, \chi_h), \quad \forall \chi_h \in S_0^h \times S_0^h. \quad (2.12)$$

Lemma 2.3. *Assume that $V(x), \phi(x) \in L^\infty(\Omega) \times L^\infty(\Omega)$, $V_1(x) \geq 0$, and $u \in (H^2(\Omega) \cap H_0^1(\Omega)) \times (H^2(\Omega) \cap H_0^1(\Omega))$. Then $P_h u$ has the error estimates:*

$$\|u - P_h u\|_s \leq Ch^{2-s}\|u\|_2, \quad s = 0, 1, \quad (2.13)$$

$$\|u - P_h u\|_{0,\infty} \leq Ch\|u\|_2. \quad (2.14)$$

Proof. From (2.12), we can get

$$\begin{aligned} \|u - P_h u\|_1^2 &\leq CA'(\phi; u - P_h u, u - P_h u) = CA'(\phi; u - P_h u, u - u_I) \\ &\leq C\|u - P_h u\|_1\|u - u_I\|_1, \end{aligned}$$

which implies that

$$\|u - P_h u\|_1 \leq C\|u - u_I\|_1. \quad (2.15)$$

It follows from (2.3) and (2.15), that

$$\|u - P_h u\|_1 \leq Ch\|u\|_2. \quad (2.16)$$

Now, we consider the auxiliary problem as follows: Find $w \in H_0^1(\Omega) \times H_0^1(\Omega)$ such that

$$\begin{aligned} a'(\phi; w, \chi) &= (\nabla w_1, \nabla \chi_1) + (V_1 w_1 + 3\phi_1^2 w_1 + \phi_2^2 w_1 + V_2 w_2 + 2\phi_1 \phi_2 w_2, \chi_1) \\ &\quad + (\nabla w_2, \nabla \chi_2) + (V_1 w_2 + \phi_1^2 w_2 + 3\phi_2^2 w_2 - V_2 w_1 + 2\phi_1 \phi_2 w_1, \chi_2) \\ &= (u_1 - P_h u_1, \chi_1) + (u_2 - P_h u_2, \chi_2), \quad \forall \chi \in H_0^1(\Omega) \times H_0^1(\Omega). \end{aligned} \quad (2.17)$$

Note that $u - P_h u \in L^2(\Omega) \times L^2(\Omega)$ similar to the proof of Lemma 2.2, the problem (2.17) has a unique solution $w \in (H^2(\Omega) \cap H_0^1(\Omega)) \times (H^2(\Omega) \cap H_0^1(\Omega))$ and

$$\|w\|_2 \leq C\|u - P_h u\|_0.$$

Let $w_I \in S_0^h \times S_0^h$ be the linear interpolation of w , and take $\chi = u - P_h u$ in (2.17). We have

$$\begin{aligned} \|u - P_h u\|_0^2 &= a'(\phi; w, u - P_h u) = A'(\phi; u - P_h u, w) \\ &= A'(\phi; u - P_h u, w - w_I) \leq C\|u - P_h u\|_1\|w - w_I\|_1 \\ &\leq Ch\|u - P_h u\|_1\|w\|_2 \leq Ch\|u - P_h u\|_1\|u - P_h u\|_0, \end{aligned}$$

which implies that

$$\|u - P_h u\|_0 \leq Ch\|u - P_h u\|_1. \quad (2.18)$$

Therefore, (2.13) follows from (2.16) and (2.18). From (2.3) and the well known inverse inequality, we can get

$$\|u - u_I\|_{0,\infty} \leq Ch\|u\|_2, \quad (2.19)$$

and

$$\begin{aligned} \|u_I - P_h u\|_{0,\infty} &\leq Ch^{-1}\|u_I - P_h u\|_0 \\ &\leq Ch^{-1}(\|u - u_I\|_0 + \|u - P_h u\|_0) \leq Ch\|u\|_2. \end{aligned} \quad (2.20)$$

Therefore, (2.14) follows from (2.19), (2.20) and the following inequality:

$$\|u - P_h u\|_{0,\infty} \leq \|u - u_I\|_{0,\infty} + \|u_I - P_h u\|_{0,\infty}.$$

This completes the lemma. $\square$

Next we introduce a Green's function. Given $z \in \Omega$, the Green's function $g^z \in H_0^1 \times H_0^1$, is defined by

$$A'(\phi; u, g^z) = \langle u, \delta^z \rangle, \quad \forall u \in H_0^1 \times H_0^1, \quad (2.21)$$

where δ^z denotes the Delta function satisfying $\langle u, \delta^z \rangle = u_1(z) + u_2(z)$.

Noticing that $a'(\phi; g^z, u) = A'(\phi; u, g^z)$, so similar to the proof of Lemma 2.2, we can get the following lemma:

Lemma 2.4. *If $V(x), \phi(x) \in L^\infty(\Omega) \times L^\infty(\Omega)$ and $V_1(x) \geq 0$, then the problem (2.21) has a unique solution $g^z \in H_0^1 \times H_0^1$.*

Therefore, we can find the discrete Green's function $g_h^z \in S_0^h \times S_0^h$ such that

$$A'(\phi; u, g_h^z) = \langle u, \delta^z \rangle, \quad \forall u \in S_0^h \times S_0^h. \quad (2.22)$$

Theorem 2.1. *Assume that*

$$\psi \in (H^2 \cap H_0^1) \times (H^2 \cap H_0^1), \quad V(x) \in L^\infty(\Omega) \times L^\infty(\Omega), \quad \text{and } V_1(x) \geq 0. \quad (2.23)$$

Then ψ_h , the finite element approximation of ψ , has the error estimate:

$$\|\psi - \psi_h\|_{0,\infty} \leq Ch. \quad (2.24)$$

Proof. From the estimate [4]

$$|v(x)| \leq C |\ln h|^{\frac{1}{2}} \|v\|_1, \quad \forall v \in S_0^h$$

and (2.22), we have

$$\|g_h^z\|_1^2 \leq C |A'(\psi; g_h^z, g_h^z)| = C |g_{h,1}^z(z) + g_{h,2}^z(z)| \leq C |\ln h|^{\frac{1}{2}} \|g_h^z\|_1,$$

which implies that

$$\|g_h^z\|_1 \leq C |\ln h|^{\frac{1}{2}}. \quad (2.25)$$

Defining a set

$$B = \left\{ v \mid v \in S_0^h \times S_0^h, \|v - P_h \psi\|_{0,\infty} \leq Ch \right\}$$

and a nonlinear operator $\Phi : B \rightarrow S_0^h \times S_0^h$ by

$$A'(\psi; \Phi(u_h), \chi_h) = A'(\psi; \psi, \chi_h) - R(\psi, u_h, \chi_h), \quad \forall \chi_h \in S_0^h \times S_0^h, \quad \forall u_h \in B. \quad (2.26)$$

Next, we verify that $\Phi(B) \subset B$ for $h \ll 1$, and Φ is continuous. From (2.12) and (2.26), we have

$$A'(\psi; \Phi(u_h), \chi_h) = A'(\psi; P_h \psi, \chi_h) - R(\psi, u_h, \chi_h), \quad \forall \chi_h \in S_0^h \times S_0^h, \forall u_h \in B. \quad (2.27)$$

Taking $\chi_h = g_h^z$ in (2.27), and from (2.4), (2.14) and (2.25), we get

$$\begin{aligned} |\Phi(u_h)(z) - P_h \psi(z)| &= |A'(\psi; \Phi(u_h) - P_h \psi, g_h^z)| = |R(\psi, u_h, g_h^z)| \\ &\leq C \left(\|u_h - \psi\|_{0,4}^2 + \|u_h - \psi\|_{0,6}^3 \right) \|g_h^z\|_0 \leq C \left(\|u_h - \psi\|_{0,\infty}^2 + \|u_h - \psi\|_{0,\infty}^3 \right) \|g_h^z\|_1 \\ &\leq C |\ln h|^{\frac{1}{2}} \left(\|u_h - P_h \psi\|_{0,\infty}^2 + \|P_h \psi - \psi\|_{0,\infty}^2 + \|u_h - P_h \psi\|_{0,\infty}^3 + \|P_h \psi - \psi\|_{0,\infty}^3 \right) \\ &\leq C |\ln h|^{\frac{1}{2}} (h^2 + h^3), \end{aligned}$$

which implies that

$$\|\Phi(u_h) - P_h\psi\|_{0,\infty} \leq Ch.$$

In other words, we have proved that $\Phi(u_h) \in B$. Moreover, for any $u_h, \tilde{u}_h \in B$, from (2.4) and (2.26), we obtain

$$\begin{aligned} & \|\Phi(u_h) - \Phi(\tilde{u}_h)\|_0^2 \leq \|\Phi(u_h) - \Phi(\tilde{u}_h)\|_1^2 \\ & \leq CA'(\psi; \Phi(u_h) - \Phi(\tilde{u}_h), \Phi(u_h) - \Phi(\tilde{u}_h)) \\ & = C\left(R(\psi, \tilde{u}_h, \Phi(u_h) - \Phi(\tilde{u}_h)) - R(\psi, u_h, \Phi(u_h) - \Phi(\tilde{u}_h))\right) \\ & \leq C\left(\|\psi - u_h\|_{0,\infty} + \|\psi - u_h\|_{0,\infty}^2 + \|\psi - \tilde{u}_h\|_{0,\infty} + \|\psi - \tilde{u}_h\|_{0,\infty}^2\right) \cdot \|u_h - \tilde{u}_h\|_0 \cdot \|\Phi(u_h) - \Phi(\tilde{u}_h)\|_0 \\ & \leq C(h + h^2)\|u_h - \tilde{u}_h\|_0 \cdot \|\Phi(u_h) - \Phi(\tilde{u}_h)\|_0. \end{aligned}$$

Therefore,

$$\|\Phi(u_h) - \Phi(\tilde{u}_h)\|_0 \leq C\|u_h - \tilde{u}_h\|_0,$$

which means that nonlinear operator Φ is continuous.

An application of Brouwer's fixed point theorem shows the existence of a $\tilde{\psi}_h \in B$ so that $\Phi(\tilde{\psi}_h) = \tilde{\psi}_h$. By definition (2.26) and (2.6), $\tilde{\psi}_h$ is the linear finite element approximation ψ_h , and satisfies

$$\|\psi_h - P_h\psi\|_{0,\infty} \leq Ch. \quad (2.28)$$

Therefore, (2.24) follows from (2.14) and (2.28). $\square$

Theorem 2.2. *Under the assumption (2.23), ψ_h has the following error estimate:*

$$\|\psi - \psi_h\|_s \leq Ch^{2-s}\|\psi\|_2, \quad s = 0, 1. \quad (2.29)$$

Proof. From (2.6) and (2.12), we have

$$A'(\psi; P_h\psi - \psi_h, \chi_h) = R(\psi, \psi_h, \chi_h), \quad \chi_h \in S_0^h \times S_0^h. \quad (2.30)$$

Taking $\chi_h = P_h\psi - \psi_h$ in (2.30), and from (2.4), we get

$$\begin{aligned} & \|P_h\psi - \psi_h\|_1^2 \\ & \leq C|A'(\psi; P_h\psi - \psi_h, P_h\psi - \psi_h)| = C|R(\psi, \psi_h, P_h\psi - \psi_h)| \\ & \leq C\left(\|\psi - \psi_h\|_{0,4}^2 + \|\psi - \psi_h\|_{0,6}^3\right)\|P_h\psi - \psi_h\|_0, \end{aligned}$$

and using (2.24), then

$$\begin{aligned} & \|P_h\psi - \psi_h\|_1 \leq C\left(\|\psi - \psi_h\|_{0,4}^2 + \|\psi - \psi_h\|_{0,6}^3\right) \\ & \leq C\left(\|\psi - \psi_h\|_{0,\infty} + \|\psi - \psi_h\|_{0,\infty}^2\right)\|\psi - \psi_h\|_0 \leq Ch\|\psi - \psi_h\|_1. \end{aligned}$$

Therefore

$$\begin{aligned} \|\psi - \psi_h\|_1 & \leq \|\psi - P_h\psi\|_1 + \|P_h\psi - \psi_h\|_1 \\ & \leq \|\psi - P_h\psi\|_1 + Ch\|\psi - \psi_h\|_1, \end{aligned} \quad (2.31)$$

From the assumption (2.23) and (2.1), the above inequality implies that

$$\|\psi - \psi_h\|_1 \leq C\|\psi - P_h\psi\|_1.$$

From (2.13), we get

$$\|\psi - \psi_h\|_1 \leq Ch\|\psi\|_2. \quad (2.32)$$

Now we consider the auxiliary problem: Find $u \in H_0^1 \times H_0^1$ such that

$$a'(\psi; u, \chi) = (\psi_1 - \psi_{h,1}, \chi_1) + (\psi_2 - \psi_{h,2}, \chi_2), \quad \forall \chi \in H_0^1 \times H_0^1. \quad (2.33)$$

Following the idea shown in the proof of Lemma 2.3, we take $\chi = \psi - \psi_h$ in (2.33), then

$$\begin{aligned} \|\psi - \psi_h\|_0^2 &= a'(\psi; u, \psi - \psi_h) = A'(\psi; \psi - \psi_h, u) \\ &= A'(\psi; \psi - \psi_h, u - u_I) + A'(\psi; \psi - \psi_h, u_I) \\ &= A'(\psi; \psi - \psi_h, u - u_I) + R(\psi, \psi_h, u_I) \\ &\leq C\left(\|\psi - \psi_h\|_1\|u - u_I\|_1 + (\|\psi - \psi_h\|_{0,4}^2 + \|\psi - \psi_h\|_{0,6}^3)\|u_I\|_0\right) \\ &\leq C\left(h\|\psi - \psi_h\|_1\|u\|_2 + h\|\psi - \psi_h\|_0\|u_I\|_0\right) \\ &\leq C\left(h\|\psi - \psi_h\|_1\|u\|_2 + h\|\psi - \psi_h\|_1(\|u\|_0 + \|u - u_I\|_0)\right) \\ &\leq Ch\|\psi - \psi_h\|_1\|u\|_2 \leq Ch\|\psi - \psi_h\|_1\|\psi - \psi_h\|_0. \end{aligned}$$

Therefore,

$$\|\psi - \psi_h\|_0 \leq Ch\|\psi - \psi_h\|_1.$$

This, together with (2.32), gives (2.29). $\square$

3. The Two-grid Finite Element Schemes

In this section, following the idea of linearization in [18, 19] and decoupling in [11], we construct the two-grid finite element algorithms for problem (1.1)-(1.2). The basic mechanisms in our approach is another finite element space $S_0^H \times S_0^H (\subset S_0^h \times S_0^h)$ defined on a coarser quasi-uniform triangulations of Ω with mesh size $H > h > 0$.

Set the bilinear forms

$$\begin{aligned} \widehat{A}'(\phi; u, w) &= (\nabla u_1, \nabla w_1) + (V_1 u_1 + 3\phi_1^2 u_1 + \phi_2^2 u_1, w_1) \\ &\quad + (\nabla u_2, \nabla w_2) + (V_1 u_2 + \phi_1^2 u_2 + 3\phi_2^2 u_2, w_2) \\ N(\phi; u, w) &= (2\phi_1 \phi_2 u_2 - V_2 u_2, w_1) + (2\phi_1 \phi_2 u_1 + V_2 u_1, w_2). \end{aligned}$$

Then it can be verified that

$$A'(\phi; u, w) = \widehat{A}'(\phi; u, w) + N(\phi; u, w). \quad (3.1)$$

Algorithm 3.1.

Step 1. Find $\psi_H \in S_0^H \times S_0^H$ such that

$$A(\psi_H, \chi_H) = (f, \chi_H), \quad \forall \chi_H \in S_0^H \times S_0^H,$$

Step 2. Find $\psi^h \in S_0^h \times S_0^h$, $\forall \chi_h \in S_0^h \times S_0^h$, such that

$$\hat{A}'(\psi_H; \psi^h, \chi_h) = A'(\psi_H; \psi_H, \chi_h) - A(\psi_H, \chi_h) + (f, \chi_h) - N(\psi_H; \psi_H, \chi_h). \quad (3.2)$$

We note that the system in Step 2 is a decoupled linear system, and only on the coarser grid the original problem needs to be solved in Step 1. As the following theorem shows, ψ^h can reach the optimal accuracy in H^1 -norm if H is taken to be $\sqrt{h}$. Because the dimension of S_0^H is much smaller than the dimension of S_0^h , the efficiency of this algorithm is evident.

Theorem 3.1. *Under the assumption (2.23), ψ^h has the following error estimate:*

$$\|\psi - \psi^h\|_1 \leq C(h + H^2).$$

Proof. From (3.1), (3.2), (2.2) and (2.5), we can easily check that

$$\begin{aligned} \hat{A}'(\psi_H; \psi_h - \psi^h, \chi_h) &= \hat{A}'(\psi_H; \psi_h, \chi_h) - \hat{A}'(\psi_H; \psi^h, \chi_h) \\ &= A'(\psi_H; \psi_h - \psi_H, \chi_h) + A(\psi_H, \chi_h) - (f, \chi_h) - N(\psi_H; \psi_h - \psi_H, \chi_h) \\ &= -R(\psi_H, \psi_h, \chi_h) - N(\psi_H; \psi_h - \psi_H, \chi_h). \end{aligned} \quad (3.3)$$

Noting that

$$\|\psi_H\|_{0,\infty} \leq \|\psi\|_{0,\infty} + \|\psi - \psi_H\|_{0,\infty},$$

we have from (2.24) that

$$\|\psi_H\|_{0,\infty} \leq \|\psi\|_{0,\infty} + CH \leq C.$$

Taking $\chi_h = \psi_h - \psi^h$ in (3.3), using (2.24) and (2.29), we have

$$\|\psi_h - \psi^h\|_1^2 \leq C \left(\|\psi_h - \psi_H\|_{0,4}^2 + \|\psi_h - \psi_H\|_{0,6}^3 + \|\psi_h - \psi_H\|_0 \right) \|\psi_h - \psi^h\|_0.$$

Then we have

$$\begin{aligned} \|\psi_h - \psi^h\|_1 &\leq C \left(\|\psi_h - \psi_H\|_{0,4}^2 + \|\psi_h - \psi_H\|_{0,6}^3 + \|\psi_h - \psi_H\|_0 \right) \\ &\leq C \left(\|\psi_h - \psi_H\|_{0,\infty} + \|\psi_h - \psi_H\|_{0,\infty}^2 + 1 \right) \|\psi_h - \psi_H\|_0 \leq CH^2. \end{aligned} \quad (3.4)$$

Therefore, from (2.29) and (3.4) we get

$$\|\psi - \psi^h\|_1 \leq \|\psi - \psi_H\|_1 + \|\psi_H - \psi^h\|_1 \leq C(h + H^2).$$

This completes the proof of the theorem. $\square$

Next, by slightly simplifying the coefficient matrix of the system in Step 2, we can get an alternative of Algorithm 3.1. Let

$$\begin{aligned} \tilde{A}(u, w) &= (\nabla u_1, \nabla w_1) + (\nabla u_2, \nabla w_2) \\ \tilde{N}(\phi; u, w) &= (V_1 u_1 + 3\phi_1^2 u_1 + \phi_2^2 u_1 - V_2 u_2 + 2\phi_1 \phi_2 u_2, w_1) \\ &\quad + (V_1 u_2 + \phi_1^2 u_2 + 3\phi_2^2 u_2 + V_2 u_1 + 2\phi_1 \phi_2 u_1, w_2). \end{aligned}$$

Then

$$A'(\phi; u, w) = \tilde{A}(u, w) + \tilde{N}(\phi; u, w). \quad (3.5)$$

Algorithm 3.2.

Step 1. Find $\psi_H \in S_0^H \times S_0^H$ such that

$$A(\psi_H, \chi_H) = (f, \chi_H), \quad \forall \chi_H \in S_0^H \times S_0^H.$$

Step 2. Find $\psi^ \in S_0^h \times S_0^h$, $\forall \chi_h \in S_0^h \times S_0^h$, such that*

$$\tilde{A}(\psi^*, \chi_h) = A'(\psi_H; \psi_H, \chi_h) - A(\psi_H, \chi_h) + (f, \chi_h) - \tilde{N}(\psi_H; \psi_H, \chi_h). \quad (3.6)$$

Theorem 3.2. *Under the assumption (2.23), ψ^* has the following error estimate:*

$$\|\psi - \psi^*\|_1 \leq C(h + H^2).$$

Proof. It follow from (3.5), (3.6), (2.2) and (2.5) that

$$\begin{aligned} \tilde{A}(\psi_h - \psi^*, \chi_h) &= \tilde{A}(\psi_h, \chi_h) - \tilde{A}(\psi^*, \chi_h) \\ &= A'(\psi_H; \psi_h - \psi_H, \chi_h) + \tilde{N}(\psi_H; \psi_H - \psi_h, \chi_h) + A(\psi_H, \chi_h) - (f, \chi_h) \\ &= -R(\psi_H, \psi_h, \chi_h) + \tilde{N}(\psi_H; \psi_H - \psi_h, \chi_h). \end{aligned} \quad (3.7)$$

Taking $\chi_h = \psi_h - \psi^*$ in (3.7) and following the idea to get (3.4), we complete the proof. $\square$

4. Numerical Example

In this section, we carry out the numerical examples to demonstrate the efficiency of our algorithms.

Example 1. For problem (1.1)-(1.2), let $V = 1+i$ and $V = \frac{1}{2}(x_1^2 + x_2^2) + i\frac{1}{2}(x_1^2 + x_2^2)$ respectively, $\Omega = [0, 1] \times [0, 1]$, and f be so chosen that

$$\psi = x_1 x_2 (1 - x_1)(1 - x_2) \exp\{x_1 + x_2\} + i \frac{x_1 x_2 (1 - x_1)(1 - x_2)}{\sin \frac{(2x_1 + 1)\pi}{4}}$$

is the exact solution.

Here Ω is uniformly divided into families T_H and T_h of triangulations, and $S_0^H, S_0^h \subset H_0^1$ are linear finite element spaces defined on T_H, T_h respectively. In our algorithms, we solve the original problem by Newton iteration method on Step 1, and solve the reduced equations by the multigrid method [14] on Step 2. For $h = H^2$ and $H = \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$, ψ^h, ψ^* and ψ_h are computed by Algorithm 3.1, Algorithm 3.2 and (2.2) respectively. From the numerical results in Table 1 and Table 2, we can see that $\|\psi - \psi_h\|_1 \approx O(h)$ and $\|\psi - \psi^h\|_1$ or $\|\psi - \psi^*\|_1 \approx O(H^2) (\approx O(h))$, which coincide with the theoretical results obtained in Theorem 2.2, Theorem 3.1 and Theorem 3.2, respectively. And, on running CPU time, we find that our two-grid finite element methods are much more efficient than the standard finite element method.

Table 1. Errors and CPU time of ψ_h , ψ^h and ψ^* , $V=1+i$.

mesh	$\ \psi - \psi_h\ _1$	r	cpu(s)	$\ \psi - \psi^h\ _1$	r	cpu(s)	$\ \psi - \psi^*\ _1$	r	cpu(s)
$h=1/16$	5.92E-2		0.79	5.80E-2		0.04	5.82E-2		0.04
$h=1/64$	1.46E-2	4.05	20.24	1.45E-2	4.00	0.60	1.46E-2	3.99	0.66
$h=1/256$	3.64E-3	4.01	1004.23	3.64 E-3	3.98	11.59	3.65E-3	4.00	12.19

Table 2. Errors and CPU time of ψ_h , ψ^h and ψ^* , $V = \frac{1}{2}(x_1^2 + x_2^2) + i\frac{1}{2}(x_1^2 + x_2^2)$.

mesh	$\ \psi - \psi_h\ _1$	r	cpu(s)	$\ \psi - \psi^h\ _1$	r	cpu(s)	$\ \psi - \psi^*\ _1$	r	cpu(s)
$h=1/16$	5.93E-2		0.81	5.80E-2		0.04	5.80E-2		0.04
$h=1/64$	1.46E-2	4.06	20.44	1.45E-2	4.00	0.59	1.45E-2	4.00	0.64
$h=1/256$	3.64E-3	4.01	1005.01	3.63E-3	3.99	11.82	3.63E-3	3.99	11.21

5. Conclusion

In this paper, we constructed and analyzed a fully discretized two-grid finite element scheme for a time-independent nonlinear Schrödinger equation. In this scheme, solution of the coupled nonlinear equations on a fine grid is approximated by the solution of the same equations on a much coarser grid together with the solution of the decoupled linear equations on a finer grid, where the key idea is to use the coarse space to produce a rough approximation of the solution, and then use it as the known-item so that the original equations are decoupled and linearized on the fine grid. Both theoretical analysis and numerical examples show that our schemes work efficiently and can reach the optimal accuracy in H^1 -norm. In future, we will work on extending this technique to more general problems such as the time-dependent nonlinear Schrödinger equation.

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References

- [1] O. Axelsson and A. Padiy, On a two level Newton type procedure applied for solving nonlinear elasticity problems, *Int. J. Numer. Meth. Engng.*, **49** (2000), 1479-1493.
- [2] J. B. Beitia, V. M. Perez-Garcia and V. Brazhnyi, Solitary waves in coupled nonlinear Schrödinger equations with spatially inhomogeneous nonlinearities, *Commun. Nonlinear Sci. Numer. Simul.*, **16** (2011), 158-172.
- [3] S.L. Chang, C.S. Chien and B.W. Jeng, An efficient algorithm for the Schrödinger-Poisson eigenvalue problem, *J. Comput. Appl. Math.*, **205** (2007), 509-532.
- [4] C. Chen and Y. Huang, High Accuracy Theory of Finite Element Methods (in chinese), Hunan Science and Thechnology Press, Changsha, China, 1995.
- [5] Y.P. Chen, L.P. Chen and X.C. Zhang, Two-grid method for nonlinear parabolic equations by expanded mixed finite element methods, *Numer. Methods Partial Differential Equations*, **29** (2013), 1238-1256.
- [6] C.S. Chien, H.T. Huang, B.W. Jeng and Z.C. Lid, Two-grid discretization schemes for nonlinear Schrödinger equations, *J. Comput. Appl. Math.*, **214** (2008), 549-571.

- [7] X. Dai, L. Shen and A. Zhou, A local computational scheme for higher order finite element eigenvalue approximations, *Inter. J. Numer. Anal. Model.*, **5** (2008), 570-589.
- [8] X. Gong, L. Shen and A. Zhou, Finite element approximations for Schrödinger equations with applications to electronic structure computations, *J. Comput. Math.*, **23** (2008), 310-327.
- [9] M.Y. He, Z.P. Huang, C. Wang and P.T. Sun, An efficient two-grid method for a two-phase mixed-domain model of polymer exchange membrane fuel cell, *Procedia Computer Science*, **9** (2012), 689-698.
- [10] L. Ignat and E. Zuazua, A two-grid approximation scheme for nonlinear Schrödinger equations: dispersive properties and convergence, *C. R. Acad. Sci. Paris, Ser. I*, **341** (2005), 381-386.
- [11] J. Jin, S. Shu and J. Xu, A two-grid discretization method for decoupling systems of partial differential equations, *Math. Comp.*, **75** (2006), 1617-1626.
- [12] X. Lu, H.W. Zhu, Z.Z. Yao, X.H. Meng, C. Zhang, C.Y. Zhang and B. Tian, Multisoliton solutions in terms of double Wronskian determinant for a generalized variable-coefficient nonlinear Schrödinger equation from plasma physics, arterial mechanics, fluid dynamics and optical communications, *Ann. Physics*, **323** (2008), 1947-1955.
- [13] D. Milovic and A. Biswas, Doubly periodic solution for nonlinear Schrödinger's equation with triple power law nonlinearity, *Int. J. nonlinear Sci.*, **7** (2009), 420-425.
- [14] S. Shu, D.D. Sun and J.C. Xu, An algebraic multigrids for higher order finite element discretizations, *Computing*, **77** (2006), 347-377.
- [15] L. Wu, Two-grid strategy for unsteady state nonlinear Schrödinger equations, *Int. J. Pure Appl. Math.*, **68** (2011), 465-475.
- [16] L. Wu, Two-grid mixed finite-element methods for nonlinear Schrödinger equations, *Numer. Methods partial differential equations*, **28** (2012), 63-73.
- [17] J. Xu, A new class of iterative methods for nonselfadjoint or indefinite problems, *SIAM J. Numer. Anal.*, **29** (1992), 303-319.
- [18] J. Xu, A novel two-grid method for semilinear equations, *SIAM J. Sci. Comput.*, **15** (1994), 231-237.
- [19] J. Xu, Two-grid discretization techniques for linear and nonlinear PDE, *SIAM J. Numer. Anal.*, **33** (1996), 1759-1777.
- [20] H.M. Zhang, J.C.Jin and J.Y.Wang, Two-grid finite-element method for the two-dimensional time-dependent Schrödinger equation, *Adv. Appl. Math. Mech.*, **5** (2013), 180-193.

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